PROGRAM 9: Resume Screening and Ranking

AIM

To screen and rank engineering resumes based on their similarity to a given job description using sentence embeddings.

ALGORITHM

1. Import the required libraries.
2. Load a pretrained sentence embedding model.
3. Define the job description.
4. Enter multiple resumes.
5. Generate embeddings for the job description and resumes.
6. Calculate cosine similarity.
7. Rank resumes based on similarity scores.

CODE

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity

model = SentenceTransformer("all-MiniLM-L6-v2")

job_description = """
We are looking for a Python developer with knowledge
of Machine Learning, SQL, Data Science and NLP.
"""

resumes = [
    "Python developer with experience in Machine Learning and SQL.",
    "Java developer with knowledge of web development and HTML.",
    "Data Science student with Python, NLP and Machine Learning skills."
]

job_embedding = model.encode([job_description])
```

```
resume_embeddings = model.encode(resumes)

scores = cosine_similarity(
    job_embedding,
    resume_embeddings
)[0]

ranking = sorted(
    zip(resumes, scores),
    key=lambda x: x[1],
    reverse=True
)

print("Resume Ranking:\n")

for i, (resume, score) in enumerate(ranking, 1):
    print("Rank:", i)
    print("Resume:", resume)
    print("Similarity Score:", round(float(score), 3))
    print()
```

INPUT

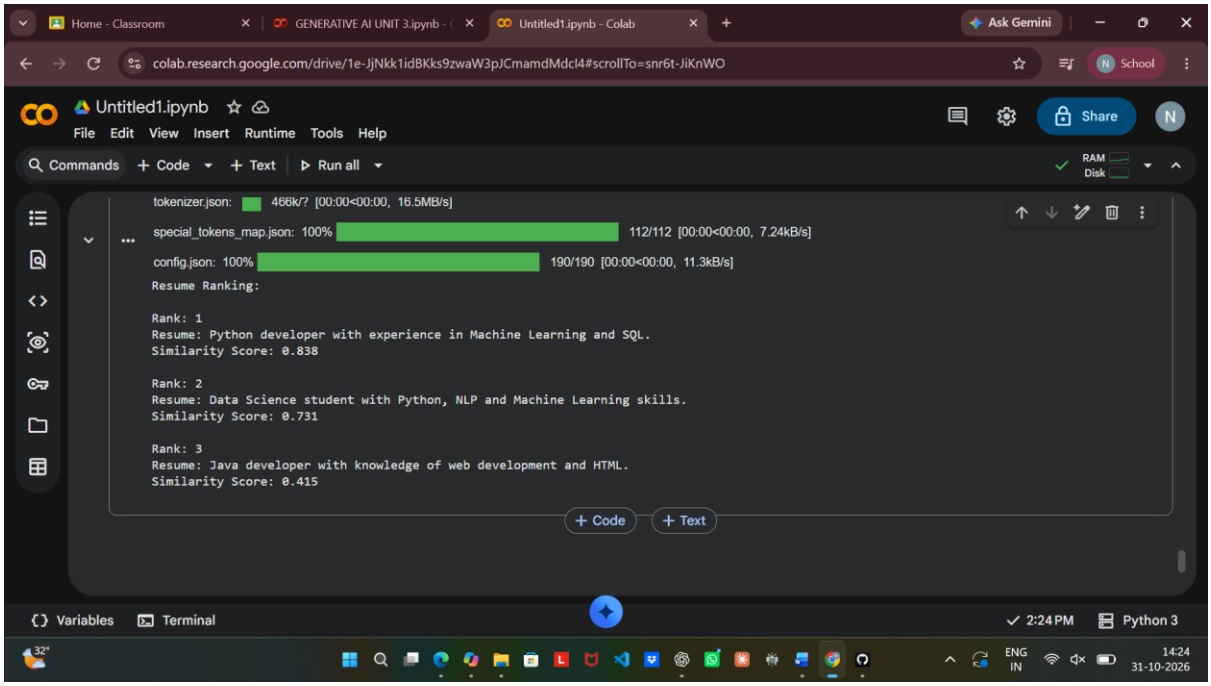
Job Description:

Python developer with knowledge of Machine Learning,

SQL, Data Science and NLP.

Resumes:

1. Python developer with Machine Learning and SQL.
2. Java developer with web development.
3. Data Science student with Python, NLP and Machine Learning.



OUTPUT

The resumes are displayed in descending order of similarity scores.

Example format:

Resume Ranking:

Rank: 1

Resume: ...

Similarity Score: ...

Rank: 2

Resume: ...

Similarity Score: ...

Exact scores and ranking depend on the embedding model.

RESULT

Thus, resumes were screened and ranked based on their semantic similarity to a job description.

Note: Similarity scores are not hiring decisions. Human review is necessary.